

A 5 MHz, 12 V, 10 A, Monolithically Integrated Two-Phase Series Capacitor Buck Converter

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Abstract—This paper presents the first monolithically integrated two-phase series capacitor buck converter. This converter achieves over 60 A/cm³ current density. The series capacitor buck converter topology enables high frequency (HF) operation up to 5 MHz per phase without special magnetics or compound semiconductors. An adaptive constant on-time controller provides fast load transient response and fixed frequency operation in steady state. The series capacitor is pre-charged before switching and monitored to protect against faults. Experimental results for a 12 V input, 1.2V/10A output application demonstrate higher peak efficiency than a conventional buck converter while operating at 4 times higher switching frequency.

Keywords—voltage regulator, switched capacitor converter, current density, series capacitor buck converter, high frequency dc-dc converter

I. INTRODUCTION

Increasing switching frequency can lead to reduced physical size of power converters. This is especially useful in point-of-load voltage regulators because they often use up considerable circuit board space. Lower values of inductance and capacitance are needed to meet the converter design parameters. Converter transient response is improved with higher switching frequencies since inductor current slew rates increase with lower inductance. There is also the added potential benefit of reducing bill of materials (BOM) costs.

The buck converter is the most commonly used converter topology for voltage regulators. When attempting to increase switching frequency into the HF range (3-30 MHz), switching losses typically becomes prohibitively large for most applications. Additionally, the on-time of the high side switch is very short (e.g. less than 30 ns) for HF, high conversion ratio (>5:1) converters. The HF buck converters on the market today are typically designed with a low input voltage (5 V or less) and low output current (1 A or less). The series capacitor buck converter, shown in Fig. 1, overcomes many challenges faced in high step-down ratio, HF buck converters [1].

Combining a switched capacitor circuit and a buck converter has several advantages. Voltage conversion can be accomplished by the switched capacitor circuit and output regulation is achieved through the buck stage. This approach plays to the strengths of each circuit. Switched capacitor circuits are typically designed for integer ratio voltage conversion (e.g. 2:1, 3:1, etc.). High voltage conversion ratios are attained with relative ease. However, output voltage regulation is realized through adjusting losses. On the other

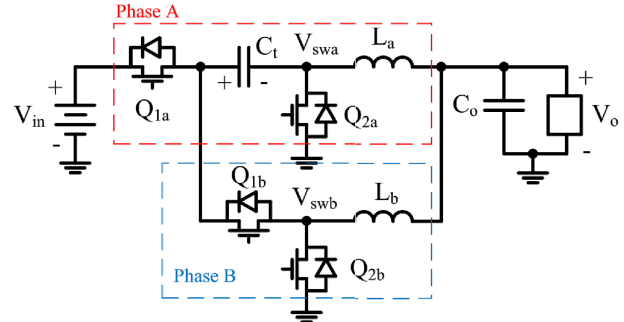


Fig. 1. The two-phase series capacitor buck converter.

hand, a buck converter can adjust its duty ratio efficiently and with high precision. The hybrid approach can be implemented in two stages consisting of a switched capacitor circuit followed by a buck converter [2], [3]. The three level buck converter combines both stages into one with soft charging of the energy transfer capacitor [4]-[7]. The merged capacitive attenuator uses a similar technique [8]-[9]. Downsides of these approaches include more components, current conduction through more than one switch at all times, and potentially a reduction in converter efficiency.

This paper introduces the first monolithically integrated series capacitor buck converter. This converter is designed to operate with up to 5 MHz switching frequency and is targeted at 12 V input and 10 A output applications. Its adaptive on-time controller provides fast transient response and fixed frequency operation in steady state. Design considerations and experimental results are included in the following sections.

II. SERIES CAPACITOR BUCK CONVERTER

The two-phase series capacitor buck converter combines a 2:1 switched capacitor converter and a two-phase buck converter. This topology is also called the double step-down buck [10] or the extended duty ratio converter [11]-[12]. It adds one capacitor (C_t) to the normal two-phase buck converter and rearranges a few connections. The series capacitor is inserted between the phase A high side and low side switches, as shown in Fig. 1. The drain of the phase B high side switch is connected to series capacitor instead of the input supply. Three or more phase versions and coupled inductor implementations have also been explored [13]-[15]. The series capacitor technique has also been applied to the tapped inductor buck converter [16].

The converter configurations and waveforms are very similar to an interleaved, two-phase buck converter. (The

reader is referred to the references for detailed analysis [1], [10], [17].) Some key differences are that the duty ratio of the high side switches is doubled, switching occurs with half the drain-to-source voltage experienced by switches in a buck converter, and inductor current balancing is automatic. Reduced voltage switching enables efficient, high frequency energy conversion. Automatic current sharing removes the need for current sensing and current sharing control loops [17]. This offers considerable benefit at high frequency.

The major limitation of the series capacitor buck converter is the voltage conversion ratio. Because the high side switches' on-times cannot overlap (i.e. 50% duty ratio limit) and because there is an inherent 2:1 step down due to the series capacitor, the minimum input voltage must be four times greater than the output voltage, in theory. When converter losses are taken in to account, the practical minimum input voltage may be five times the output voltage. There are point-of-load applications where this limitation is not acceptable; other high conversion ratio applications may not have any problem with this constraint. Another downside to the series capacitor buck converter is that phase shedding and adding is not feasible. Other light load modes using discontinuous conduction mode and pulse frequency modulation can accommodate the need for efficient light load operation [18].

III. INTEGRATED CIRCUIT OVERVIEW

A monolithic integrated circuit (IC) implementation of the series capacitor buck converter was developed for low voltage point-of-load applications (e.g. less than 2 V) powered by a 12 V supply rail. An overview of the IC is shown in Fig. 2. The controller, gate drivers, power MOSFETs, internal regulators, and supporting circuitry are included. Passive components like input and output capacitors, inductors, feedback resistors, etc. are located externally. The chip includes typical protection circuits such as input under/overvoltage lockout, output under/overvoltage protection and overcurrent protection. New features were included to manage the series capacitor voltage. The converter was designed to operate with 2 MHz to 5 MHz per phase switching frequency, 8 V to 14 V input, and up to 10 A output. Several important typical converter parameters are listed in Table 1 for a 2 MHz and a 5 MHz design. The case

TABLE 1. TYPICAL CONVERTER PARAMETERS

Component	Parameter	
Switching frequency (f_s)	2 MHz	5 MHz
Inductance (L)	220 nH (1210)	100 nH (1210)
Inductor DCR	10 m Ω	5 m Ω
Series capacitance (C_t)	2.2 μ F (1206)	1 μ F (1206)
Output capacitance (C_o)	147 μ F	91 μ F
$R_{ds,on}$: Q_{1a} , Q_{1b}	27 m Ω	
$R_{ds,on}$: Q_{2a} , Q_{2b}	6.8 m Ω , 9.3 m Ω	

size for the inductors and series capacitor is included using the imperial code.

A. Gate Drive Scheme

The gate driver design is an important practical consideration. Typical, ground referenced low side gate drivers powered from the VG+/- pins were used for the low side MOSFETs. The gate drive supply rail (VG+/-) is supplied by an internal linear regulator or can be supplied externally. The phase B high side MOSFET driver utilized a bootstrap capacitor to bias its floating gate drive circuit.

A unique approach was required to drive the phase A high side MOSFET because its source is floating on top of the series capacitor. A linear regulator generates a supply rail on the VGA pin. The target voltage is dynamically adjusted based on the VG+ voltage and series capacitor voltage. The linear regulator is fairly efficient because the VGA voltage is a small amount lower than the input supply (e.g. 11 V from a 12 V input). The phase A bootstrap capacitor is charged from the VGA rail when the phase A low side MOSFET is on.

B. Protection Circuits

The IC includes several protection circuits. Inductor currents, which indirectly represent the output current, are monitored using the low side MOSFETs. If the measured current exceeds predefined thresholds for three switching cycles in a row, an overcurrent fault is flagged and the converter shuts down. After a wait time of at least 10 ms, the converter restarts automatically in a scheme commonly referred to as a "hiccup" restart.

The input voltage and output voltage is also monitored. If the input voltage falls below 7.65 V or rises above 15.4 V, the converter will shut down and restart when the input voltage is back in bounds. The output voltage is protected in a similar manner. If the output is greater than $\pm 10\%$ away from the target voltage, the converter will shut down and restart after the hiccup wait time has expired.

IV. SERIES CAPACITOR DESIGN AND MANAGEMENT

The series capacitor is a critical component of the converter. Proper capacitor selection and management during operation is important.

A. Capacitor Selection

A major function of the series capacitor is energy transfer. This is a different role from input and output capacitors used in buck converters where decoupling is the primary function. In

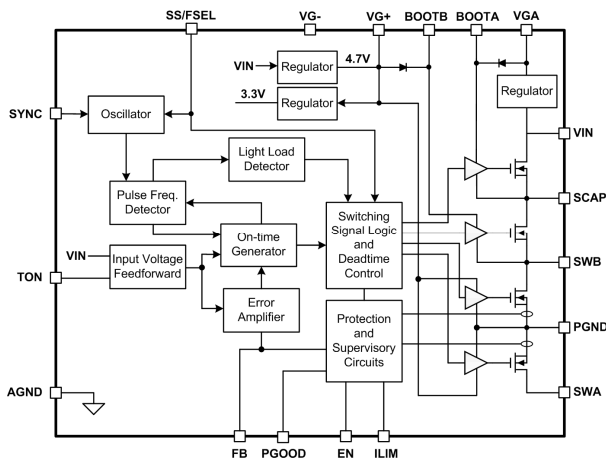


Fig. 2. Integrated series capacitor buck converter block diagram.

many ways, the series capacitor is similar to the capacitor used for energy transfer in SEPIC converters and can be designed accordingly. A design objective may be to ensure the series capacitor voltage ripple does not exceed 5% of the nominal voltage under the worst case conditions. The equation that describes the series capacitor voltage ripple is

$$\Delta V_{Ct} = \frac{DTI_{Ct}}{C_t} = \frac{\left(\frac{2V_o}{V_{in}}\right)\left(\frac{I_o}{2}\right)}{C_t f_{sw}} \quad (1)$$

where D is the duty ratio, T is the switching period, I_{Ct} is the series capacitor current, C_t is the series capacitance, V_o is the output voltage, V_{in} is the input voltage, I_o is the output current, and f_{sw} is the switching frequency. Equation (1) can be rearranged to provide the design equation for series capacitor selection which is

$$C_t \geq \frac{\left(\frac{2V_o}{V_{in}}\right)\left(\frac{I_o}{2}\right)}{k_{Ct} \left(\frac{V_{in}}{2}\right) f_{sw}} \quad (2)$$

where k_{Ct} represents the voltage ripple percentage and $V_{in}/2$ is the nominal series capacitor voltage. If the ripple target is 5%, the value for k_{Ct} is 0.05. The largest voltage ripple occurs at full load current (highest I_o), highest duty ratio (lowest input voltage/highest output voltage), and lowest frequency.

Another aspect to consider is capacitor RMS current rating. This impacts the temperature rise of the capacitor. If the temperature rise is too large for a single capacitor, multiple capacitors may be placed in parallel. The series capacitor has the same current profile as the high side MOSFETs. The RMS current squared can be expressed as

$$I_{Ct,RMS}^2 = 2DI_{L,RMS}^2 \quad (3)$$

where $I_{L,RMS}$ is the RMS inductor current of either inductor. The series capacitor RMS current can be expressed as

$$I_{Ct,RMS} = \sqrt{2 \left(\frac{2V_o}{V_{in}}\right) \left[\left(\frac{I_o}{2}\right)^2 + \left(\frac{\Delta I_L}{12}\right)^2 \right]} \quad (4)$$

where ΔI_L is the inductor current ripple. The largest RMS current occurs at the highest load current and highest duty ratio.

Multilayer ceramic capacitors (MLCC) are well suited for operating as the series capacitor. The equivalent series resistance (ESR) is relatively low (e.g. 5 mΩ to 10 mΩ) which helps to reduce power loss and self heating. The equivalent series inductance (ESL) is fairly low which results in a high self resonant frequency (SRF). There are a few key items that should be considered when designing. First, the effective capacitance decreases with dc bias. This means that the capacitor should be selected based on its capacitance with the nominal voltage of $V_{in}/2$ applied. Temperature variation also reduces effective capacitance. For this reason, X7R capacitors with up to 125°C operating temperature range are

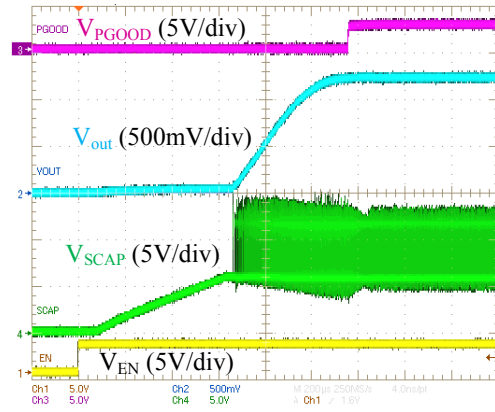


Fig. 3. Startup begins with series capacitor pre-charge (200 μs/div).

recommended. If capacitors are not properly selected, cracking or other failure modes may result.

B. Capacitor Pre-Charge

The converter pre-charges the series capacitor to half the input voltage before switching begins. This ensures predictable, smooth startup and automatic, even current sharing [6], [17]. Since the series capacitor voltage is nominally $V_{in}/2$ during normal operation, that target voltage was selected for the pre-charge circuit. A 10 mA current source charges the series capacitor. The startup delay due to pre-charging the series capacitor can be calculated to be

$$t_{pc} = \frac{C_t \left(\frac{V_{in}}{2}\right)}{I_{pc}} \quad (5)$$

where t_{pc} is the series capacitor pre-charge time and I_{pc} is the pre-charge current (10 mA in this converter). An experimental result demonstrating the series capacitor pre-charge is shown in Fig. 3 for a 1 μF series capacitance. The pre-charge time is approximately 600 μs. This delay can complicate startup tracking functionality. It also highlights a tradeoff with series capacitor selection. A larger series capacitance will result in a longer pre-charge delay.

C. Voltage Monitoring

A unique function included in the IC is series capacitor voltage monitoring. The differential voltage across the series capacitor is continuously measured. A fault is triggered if the voltage strays away from the nominal voltage by more than ±30%. If the voltage is too low, the converter shuts down, waits for a hiccup restart time, pre-charges the series capacitor to half the input voltage, and initiates soft start. If the voltage is too high, the same response occurs except an internal bleed resistor is applied to reduce the series capacitor voltage. This functionality is included to prevent any damage that could be caused by series capacitor voltage going out of bounds.

V. ADAPTIVE ON-TIME CONTROL

An adaptive on-time control scheme is implemented as shown in Fig. 4. Its core operates in much the same way that constant on-time controllers work. The output feedback is compared to an internal voltage reference to trigger a high side switch on-time. This approach provides fast response to load

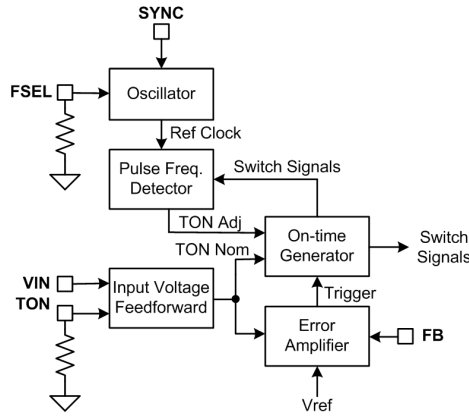


Fig. 4. Adaptive on-time controller block diagram.

transients. When the load steps up, the controller quickly fires successive on-time pulses to increase inductor current. When the load steps down, the high side switches remain off, and the inductor currents fall.

A distinguishing feature of this controller is that the on-time is adapted for fixed frequency operation in steady state. This applies over all variations in the input voltage, load current, and temperature. The nominal on-time is set by a programming resistor and an input voltage feedforward circuit. This is similar to other constant on-time controllers where the on-time is adjusted based on the input voltage. In this controller, as shown in Fig. 4, the on-time is adapted by feeding in a correction term that slowly increases or decreases the on-time based on the output of the pulse frequency detector. The pulse frequency detector compares the on-time pulse train to a reference clock derived from either an internal oscillator or an external synchronization clock signal. This forms a phase lock loop (PLL).

The PLL aligns the frequency and phase of the on-time pulses with the reference clock. When the external synchronization clock is applied to the SYNC pin, it overrides the internal clock. When the external clock is removed, the internal clock will take over after a few switching cycles. Both internal and external clocks are two times the per phase switching frequency because the controller is operating at twice the switching frequency.

The PLL bandwidth is about ten times lower in frequency than the converter closed-loop crossover frequency. This enables the converter to quickly respond to fast, large signal load transients and operate with fixed frequency in steady-state. It keeps the frequency from sliding into unwanted frequency bands. It also allows multiple converters operating from the same input bus to have their on-times interleaved to reduce voltage ripple on the input bus. This is not possible with conventional constant on-time controllers.

VI. EXPERIMENTAL RESULTS

The series capacitor buck converter was tested over various operating conditions. The typical converter parameters are listed in Table 1. A picture of the converter prototype used in testing is shown in Fig. 5. A state-of-the-art 10 A point-of-load buck converter [19] is shown in Fig. 6 for comparison. The

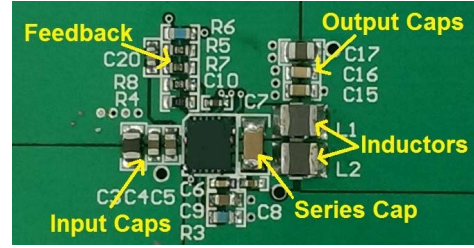


Fig. 5. 10 A, 2-5 MHz per phase series capacitor buck converter prototype.

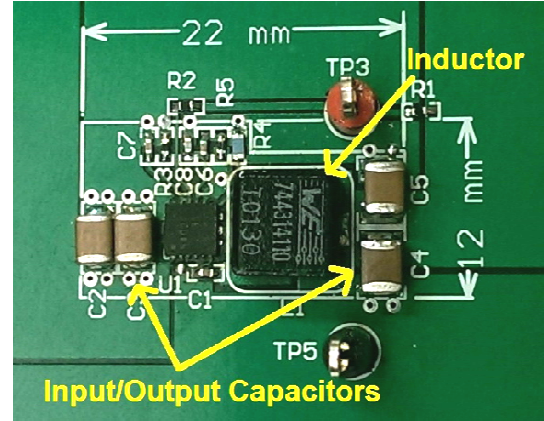


Fig. 6. 10 A, 500 kHz buck converter reference design.

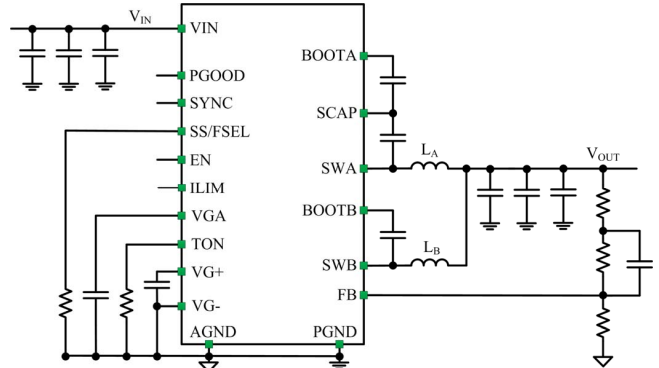


Fig. 7. A simplified schematic of the series capacitor buck converter prototype.

current density of the series capacitor buck converter is improved by use of small inductors that are 3.2 x 2.5 x 1.2 mm as compared with the 500 kHz, 10 A buck converter's inductor that is 7 x 7 x 4 mm. A major benefit is height reduction. Because it is only 1.2 mm tall, the 10 A converter can be placed in low profile locations such as on the back side of the main circuit board or on a plug in card (e.g. DIMM, PCIe). A schematic of the series capacitor buck converter prototype is shown in Fig. 7. Some additional board components used for testing the converter are not shown for simplicity. An additional 50 Ω resistor is included in the feedback network to aid with injecting a perturbation signal for closed loop measurements. A capacitor is placed in parallel with the upper feedback resistor to provide extra phase boost at the crossover frequency. The prototype was designed with a four layer PCB for ease of routing signals and to have sufficient ground planes for ground return current as well as heat removal. The IC uses flip chip, HotRod QFN packaging.

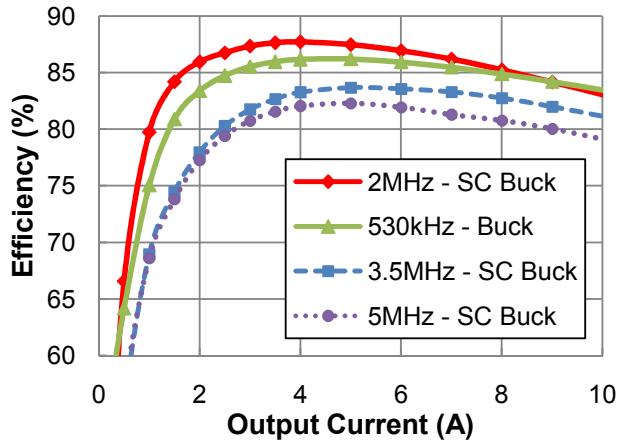


Fig. 8. The peak efficiency of the series cap buck at 2 MHz is higher than a conventional buck at 530 kHz for a 12 V input, 1.2 V output application.

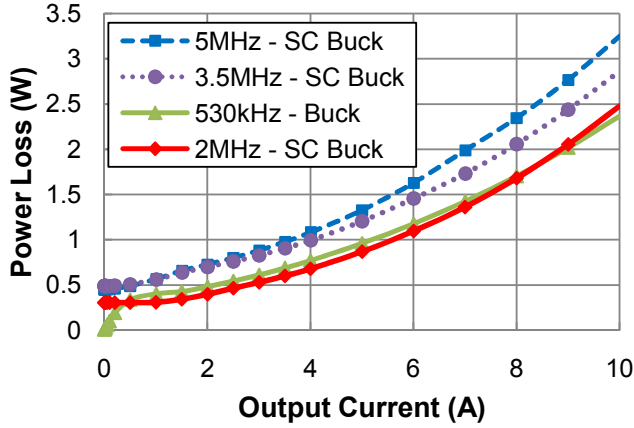


Fig. 9. Power loss comparison of the series cap buck and a standard buck for 12 V input, 1.2 V output application.

A. Efficiency and Power Loss

The measured efficiency and power loss for 12 V input, 1.2 V output, at room temperature are shown in Figs. 8 and 9. The feedback resistors and frequency setting resistor were adjusted for the buck converter reference design [19] to set its output to 1.2 V and switching frequency to approximately 500 kHz. The efficiency data collected includes all power stage, inductors, gate drive, controller, and layout losses. The only additional loss term the buck converter had was an internal linear regulator that provided power from the 12 V input to the 5 V gate driver. An external supply provided the gate drive power for the series capacitor buck converter.

The series capacitor buck converter demonstrates higher peak efficiency (87.7%) at 2 MHz per phase switching frequency than a buck converter operating at 530 kHz switching frequency (86.2%). The lower full load efficiency is attributed primarily to the higher inductor DCR (10 mΩ in the series cap buck vs. 3.2 mΩ in the buck converter). As expected, the 3.5 MHz and 5 MHz efficiency of the series capacitor buck converter is lower but still above 80% for a large portion of the load range.

The power loss results in Fig. 9 are for the same conditions as the efficiency results in Fig. 8. The series capacitor buck converter was operated in forced continuous conduction mode

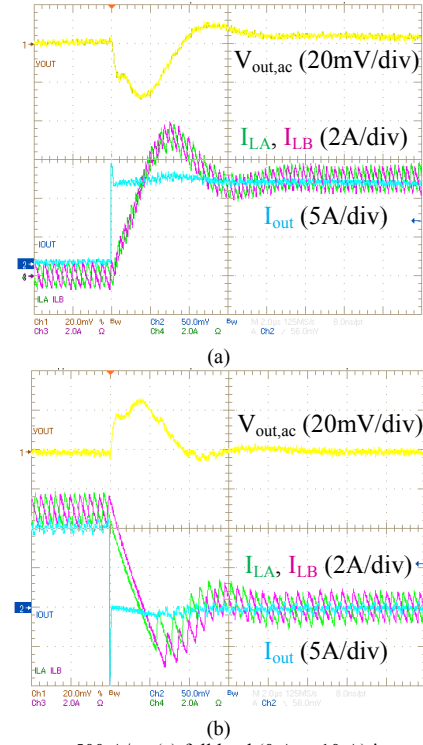


Fig. 10. Response to 500 A/μs (a) full load (0 A to 10 A) increase and (b) full load (10 A to 0 A) decrease (2 μs/div).

(FCCM) over the entire load range. The buck converter operated with pulse skipping at light loads that helped to reduce power consumption below 0.5 A output. It is also worth noting that more effort was taken to select the optimal inductor for use at 2 MHz in the series capacitor buck converter; the 3.5 MHz and 5 MHz cases used a different inductor vendor.

B. Large Signal Load Transient Response

The 2 MHz converter response to a 500 A/μs full load increase and decrease is shown in Fig. 10. The output recovery time is 4 μs with peak voltage excursion of approximately ±25 mV (i.e. ±2%). The output capacitance was increased for this test to approximately 350 μF. The fast transient load was implemented with an on-board MOSFET load. The inductor currents are well balanced even during the large, fast transient. The inductor currents are slightly imbalanced at the end of the load step down due to the controller response but quickly come back to even current sharing. Faster load transient response is possible when operating at 5 MHz per phase.

C. Closed Loop Response

The bode plot measured for the 5 MHz series capacitor buck converter is shown in Fig. 11 for two different values of output capacitance. The converter demonstrates a 219 kHz crossover frequency with 62.6 degrees of phase margin with 138 μF of output capacitance. The crossover frequency increases to 320 kHz with 53.8 degrees of phase margin when the output capacitance is reduced to 91 μF. Both scenarios were tested for 12 V input and 1.2 V output with a power resistor load drawing approximately 2.5 A. The results demonstrate stability of the system with high closed loop bandwidth. Most buck converters designed for similar

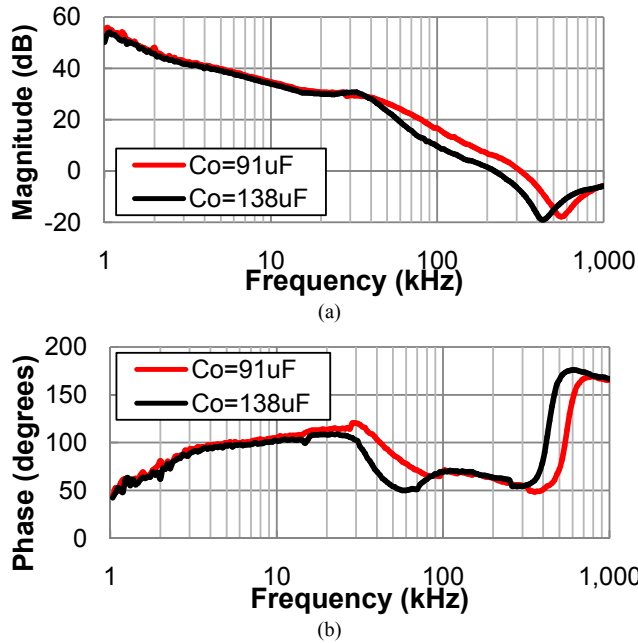


Fig. 11. Bode plot showing (a) magnitude and (b) phase for 12 V in, 1.2 V/2.5 A out at 5 MHz for two output capacitance values.

applications have closed loop bandwidths in the tens of kHz range. Extra poles and zeroes can be inferred from the gain and phase plots in the 300 kHz to 700 kHz range. The “wiggling” in the plot is attributed to higher order parasitics in the converter layout and components.

D. Thermal Image

A thermal image of the converter prototype operating at 2 MHz per phase with 12 V input, 1.2V/10A output is shown in Fig. 12. The bar on the right side of the image displays the thermal mapping with temperature measured in degrees Celsius. No air flow was applied and the ambient condition was room temperature. The result indicates that the IC and the series capacitor are the warmest components, and the hottest part of the image reaches 67°C. Most of the power loss is in the IC (more than half the total power loss) so it is understandable that it would be warm. The series capacitor has relatively low power loss but its heat sinking is not very good. The body of the series capacitor is red indicating a high temperature but the ends of the capacitor that are soldered to the PCB are green indicating a much cooler temperature.

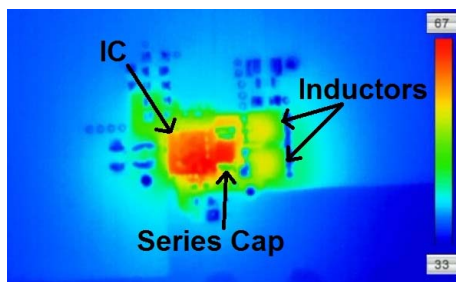


Fig. 12. Thermal image of converter prototype operating at 12 V input, 1.2V/10 A output, 2 MHz per phase switching frequency.

The inductors are fairly cool by comparison. This is noteworthy because most of the overall converter size reduction comes from using smaller inductors. There does not

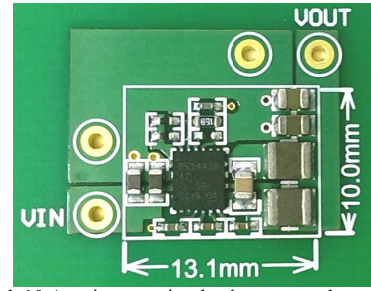


Fig. 13. Practical, 10 A series capacitor buck converter layout with small size.

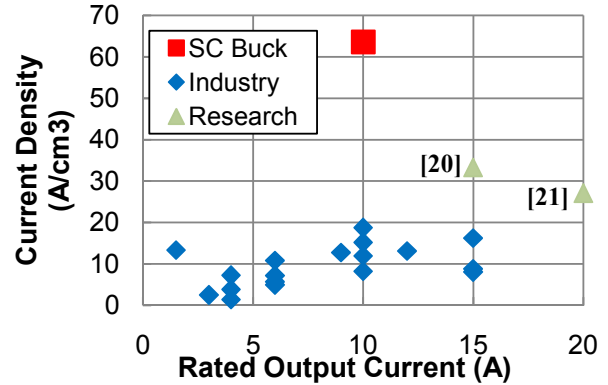


Fig. 14. The integrated series cap buck converter demonstrates several times higher current density than industry state-of-the-art and other research.

appear to be any extra thermal dissipation burden or hot spot when using the small inductors.

VII. SIZE COMPARISON

The current density of the series capacitor buck converter is compared to state-of-the-art buck converters available in industry today as well as a couple research prototypes [20], [21]. A practical, compact layout of the series capacitor buck converter without the extra components used for testing is shown in Fig. 13. The converter footprint is 131 mm² and the volume, not including the printed circuit board (PCB) thickness, is 157 mm³. The resulting current density is 63.6 A/cm³.

All the converters compared in Fig. 14 are designed for 12 V input, point-of-load applications. The total solution size is estimated including input and output capacitors, inductors, feedback network, etc. The PCB thickness is also included for the research prototypes [20], [21] since the inductor is embedded in the PCB substrate. It is clear from the figure that the series capacitor buck converter has significantly higher current density (over 60 A/cm³) than any of the conventional buck converters.

Current density is a more appropriate metric for comparing the size of point-of-load dc-dc converters than power density. Power density of a converter is often calculated at the maximum output voltage and output current. This is not useful information for point-of-load converters where the output voltage in a typical application is much lower (e.g. 1.2 V) than the maximum output voltage (e.g. 90% or more of the input voltage). Power density at a specific output voltage of interest could be calculated; however, this ends up being a scaled version of current density.

VIII. CONCLUSIONS

This paper presents a 2 MHz to 5 MHz, 12 V input, 10 A output, integrated series capacitor buck converter. The converter benefits from lower switching loss, automatic current balancing, and on-time doubling inherent to the series cap buck topology. The IC implementation includes the controller, gate drivers, power MOSFETs, and support circuitry. The series capacitor pre-charge circuit and voltage monitoring are new functions integrated in this converter. The series capacitor design equations are included for reference.

Experimental results for a 12 V to 1.2 V application demonstrate the effectiveness of the series capacitor buck converter. The converter has higher efficiency (87.7% peak) operating at 2 MHz per phase than a conventional buck converter (86.2% peak) operating at 530 kHz. An adaptive constant on-time control scheme is implemented with fast transient response achieving 4 μ s recovery time to full load steps (0 A to 10 A) slewing at 500 A/ μ s. Fixed frequency operation in steady state is also achieved. High closed loop crossover frequency of over 300 kHz was measured with over 50 degrees of phase margin. This converter including all passive components in a practical layout has the highest known current density for a 12 V input, 10 A output point-of-load dc-dc converter.

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